

BIOL-1 Exam 03 Answer Key

Part A: Multiple Choice (30 points)

1 point each.

Module 12: Darwin and Evolution

1. **C** (Shared ancestors; heritable change over generations)
2. **A** (Relative genetic contribution / reproductive success → next generation)
3. **B** (Overproduction of offspring)
4. **B** (Shared structure, modified function → homology)
5. **C** (Convergent evolution / similar selection pressures)
6. **B** (Allele-frequency change in populations across generations)
7. **B** (Acclimation within life vs heritable evolution across generations)
8. **B** (Biogeography paired with evolutionary reasoning)

Module 13: How Populations Evolve

1. **B** (Allele-frequency change over time)
2. **C** (Gene flow)
3. **B** (Bottleneck / drift from random catastrophe)
4. **B** (Founder effect)
5. **C** (Stabilizing selection)
6. **C** (Disruptive selection)
7. **B** (Mathematical baseline / null model)
8. **B** (Mechanisms that can change allele frequencies)

Module 14: Macroevolution

1. **C** (Biological species concept / isolation + interbreeding)
2. **B** (Prezygotic temporal isolation)
3. **B** (Postzygotic reduced hybrid viability)
4. **C** (Postzygotic reduced hybrid fertility)

5. **B** (Allopatric speciation)
6. **B** (Sympatric speciation via polyploidy)
7. **B** (Macroevolution as accumulated population-level processes)

Module 15: Population and Systems Ecology

1. **D** (Exponential / J-shaped phase)
 2. **A** (Carrying capacity)
 3. **C** (Density-dependent limitation)
 4. **D** (Density-independent disturbance in this idealization)
 5. **A** (More primary production needed with more trophic transfers / ~10% rule)
 6. **C** (Energy dissipated largely as heat between trophic transfers)
 7. **D** (Decomposers in nutrient cycling)
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Part B: Fill in the Blank (11 points)

1 point each. Minor spelling errors are acceptable when intent is clear.

1. **Allele frequencies** (accept allele frequency)
 2. **Gene flow**
 3. **Genetic drift**
 4. **Homologous**
 5. **Allopatric**
 6. **Sympatric**
 7. **Prezygotic** (accept pre-zygotic)
 8. **Postzygotic** (accept post-zygotic)
 9. **Hardy–Weinberg** (accept Hardy Weinberg)
 10. **Directional selection**
 11. **Decomposers**
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Part C: Free Response (9 points)

Choose *THREE* out of five. Each question is worth 3 points.

Award partial credit as appropriate.

1. Natural Selection and Evidence.

- *Natural selection*: heritable **variation**; **more offspring produced than survive/reproduce** (overproduction); **unequal survival/reproduction** tied to traits → changes in allele frequencies when heritable.
- *Evidence*: **fossils** (change over time); **homology** (shared structure, different function); **biogeography** (related forms in regionally sensible patterns); **molecular** similarities—student names **one** pattern clearly.

2. Drift, Flow, and Selection.

- **Drift**: random sampling of alleles; strongest in **small** populations; **not** “for” adaptive fit by itself.
- **Selection**: consistent differences in survival/reproduction tied to **heritable** traits → adaptive directional/stabilizing/disruptive patterns.
- **Bottleneck**: catastrophe randomly wipes most individuals; survivors may have skewed allele frequencies.
- **Founder**: small group starts new population; gene pool reflects founders’ alleles.
- **Gene flow**: alleles move **between** populations via migration/mating—can increase similarity or introduce variation.

3. Reproductive Isolation and Speciation.

- **Reproductive isolation**: reduced or blocked **gene flow** between groups (biological species framing: viable fertile offspring within group vs barriers **between** groups).
- **Prezygotic example**: temporal isolation, behavioral differences, habitat separation, mechanical mismatch, gametic incompatibility—any valid example.
- **Postzygotic example**: hybrid inviability, hybrid sterility (mule), hybrid breakdown—any valid example.

- **Allopatric:** geographic separation first, then divergence. **Sympatric:** divergence **without** geographic isolation (e.g., polyploid plants).

4. Population Growth and Regulation.

- **Exponential:** rapid increase when resources abundant (J curve). **Logistic:** growth slows as resources tighten; approaches **K** (S curve).
- **K: carrying capacity**—environment-limited population size over time.
- **Density-dependent:** disease/contacts/competition intensify with crowding—student gives plausible example.
- **Density-independent:** fire/freeze/storm kills regardless of density (idealized)—student gives plausible example.

5. Energy and Nutrients.

- **Nutrients** cycle (producers ↔ consumers ↔ **decomposers**); **energy** enters as sunlight (photosynthesis), passes through trophic levels, and leaves largely as **heat**—not recycled like atoms.
- **~10% rule:** roughly ~10% of energy passes each trophic step; long chains lose repeated fractions at each transfer, so supporting carnivores/tuna diets needs **more** net primary production than eating plants directly for the same usable human calories.

End of exam key.